JSON-Storm: prompt sweep findings

Complex batches are harder to decode reliably

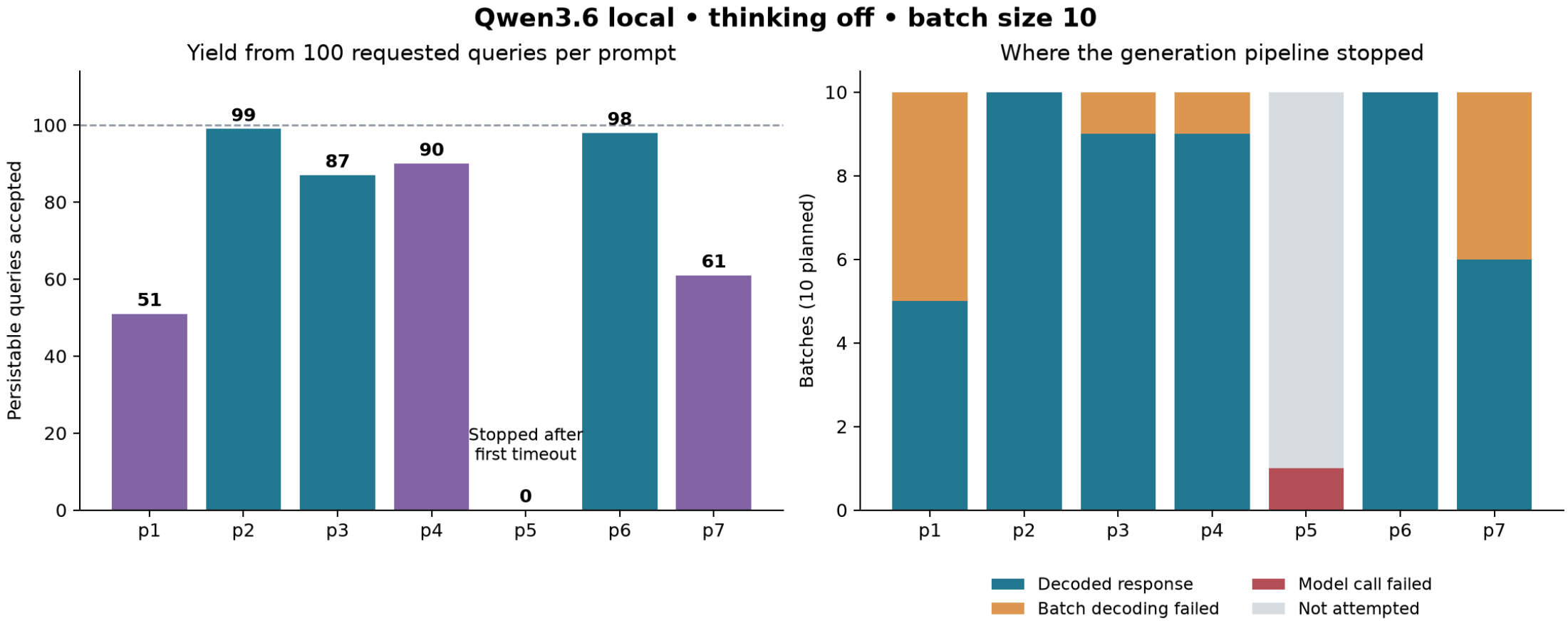
Qwen3.6 via Ollama · thinking off · batch size 10

Seven prompts × 100 requested queries, using the `mathexchange` dataset.

486 queries accepted across six completed runs.

One run stopped at its first model timeout.

Query yield depends strongly on the task



p2/p6: 98–99 accepted. Complex aggregations p1/p7: 51–61.

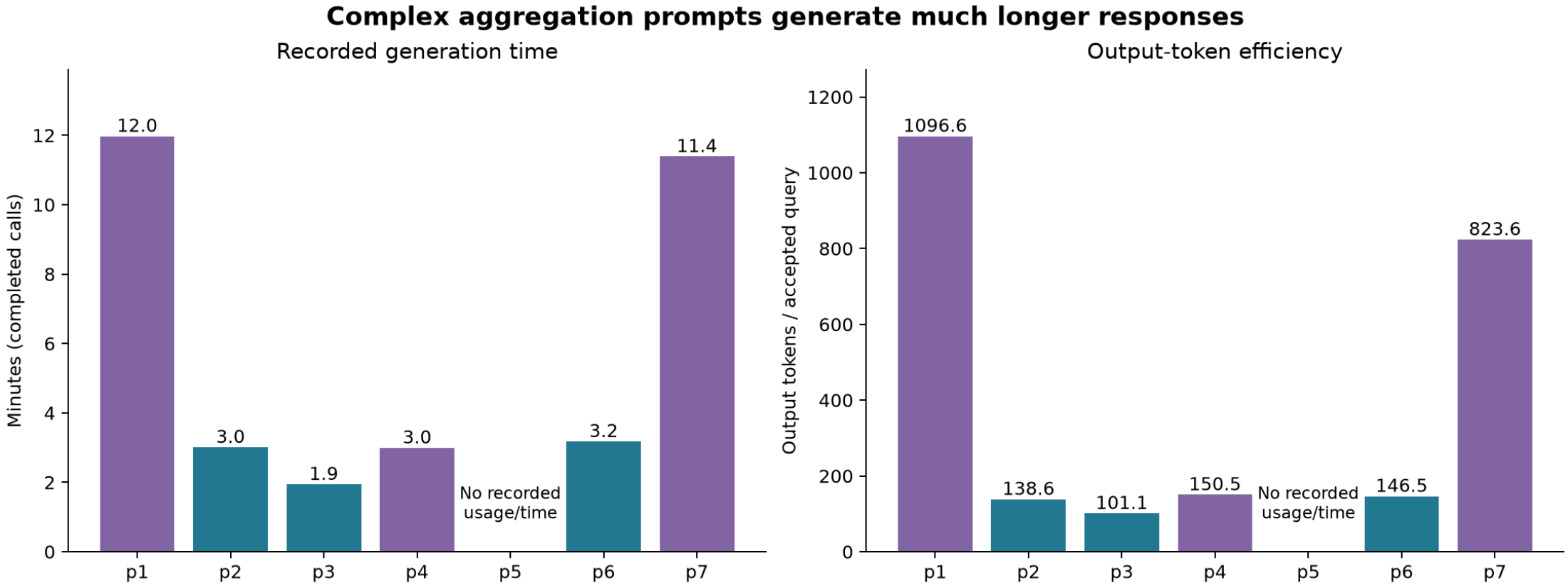
Most losses happen before individual validation

Failure	Where	Impact
Malformed JSON / duplicate keys	p1, p4, p7	11 entire batches rejected
Shell `ObjectId(...)` syntax	p3	1 entire batch rejected
`skip: null`	p2, p3, p6	6 individual queries rejected
Model-call timeout	p5	1 failed call; 9 batches unattempted

One malformed fragment can lose a ten-query batch.

The 12 rejected batches covered 120 requested query slots.

Complex output also costs more time and tokens



p1/p7 took **~12 minutes**, versus **~2–3 minutes** for simpler tasks.

What we can—and cannot—conclude

- **Batch decoding is the main bottleneck** for long aggregation output.
- **Extended JSON is working:** no persistence failures in this sweep.
- Accepted queries still need MongoDB syntax, result-count and timeout checks.
- Each run sampled a **different schema snapshot**; comparison is not fully controlled.
- One run per prompt is exploratory evidence, not a reliable ranking.

p5 needs a rerun: its first call exceeded the configured 120-second deadline.

Next: improve yield without hiding failures

1. **Fix the input:** reuse one saved schema/sample snapshot across runs.
2. **Sweep batch size:** 1, 2, 5, 10—especially for p1 and p7.
3. **Recover p5:** test a smaller batch or a separately recorded longer timeout.
4. **Clarify examples:** `skip: 0`, Extended JSON ObjectIds, combined date bounds.
5. **Test structured output separately:** compare JSON mode against the baseline.

Retain raw failures; repeat runs and measure yield, diversity, time and tokens.

Evidence and reproducibility

Prompt	Experiment	Prompt	Experiment
p1	`77f31e59`	p5	`c1da7a1f`
p2	`072f15a0`	p6	`cb4bfa15`
p3	`c5c119de`	p7	`f97b3445`
p4	`0fde4fd3`		

Analysis notes · Summary data · Batch data · Failure details

Rebuild figures from the repository root:

```
uv run python results/prompt-sweep-2026-11-03/analyze.py
```